Page-rank, centrality, and related topics

- Google's Page Rank
- Kleinberg's Hubs and Authorities
- Centrality in graphs

Google's Page rank

- Idea is to put order into the web by ranking pages by their importance..
- This idea was at the origin of Google !
- Still a lot of mystery in what is in google rank
- Model has been extended to other tasks, e.g., ranking researchers by their citations.

Page-rank

Main point: A page is important if it is pointed to by other important pages.

- Can be viewed as an application of Markov Chains
- Importance of your page (its PageRank) is determined by summing the page ranks of all pages which point to it.
- Weighting: If a page points to several other pages, then the weighting should be distributed proportionally.
- Imagine many tokens doing a random walk on this graph:
  - (δ/n) chance to follow one of the n links on a page,
  - $(1-\delta)$ chance to jump to a random page.
  - What's the chance a token will land on each page?

Page-Rank - definitions

If T_1, ..., T_n point to page T_i then $\rightarrow$

$$\rho(T_i) = 1 - \delta + \delta \left[\frac{\rho(T_1)}{|T_1|} + \frac{\rho(T_2)}{|T_2|} + \cdots \frac{\rho(T_n)}{|T_n|} \right]$$

- $|T_j|$ = count of links going out of Page T_j. So the 'vote' $\rho(T_j)$ is spread evenly among $|T_j|$ links.
- Sum of all PageRanks == 1: $\Sigma_T \rho(T) = 1$
- δ is a 'damping' parameter close to 1 – e.g. 0.85

✦ Defines a (possibly huge) Hyper-link matrix H

$$h_{ij} = \begin{cases} \frac{1}{|T_i|} & \text{if } i \text{ points to } j \\ 0 & \text{otherwise} \end{cases}$$

✍ 1 4 Nodes

A points to B and D

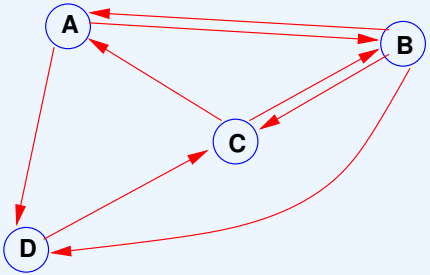
B points to A, C, and D

C points to A and B

D points to C

1) What is the H matrix?

2) The graph?



	A	B	C	D
A		$1/2$		$1/2$
B	$1/3$		$1/3$	$1/3$
C	$1/2$	$1/2$		
D			1	

- Row- sums of H are = 1.
- Sum of all PageRanks will be one:

$$\sum_{\text{All-Pages}_A} \rho(A) = 1.$$

- H is a stochastic matrix [actually it is forced to be by changing zero rows]

Algorithm: (PageRank)

1. Select initial row vector v ($v \geq 0$)
2. For i=1:maxitr
3. $\quad v := (1-\delta)e^T + \delta vH$
4. end

✍ 2 Do a few steps of this algorithm for previous example with $\delta = 0.85$.

✦ This is a row iteration..

$$\boxed{v} = \boxed{(1-\delta)e^T} + \boxed{v}\,.\,\boxed{\delta H}$$

A few properties:

- v will remain ≥ 0. [combines non-negative vectors]
- More general iteration is of the form :

$$v := v\underbrace{[(1-\delta)E + \delta H]}_{G} \quad \text{with} \quad E = ez^T$$

where z is a probability vector $e^T z = 1$
[Ex. $z = \frac{1}{n}e$]

- A variant of the power method.
- e is a right-eigenvector of G associated with $\lambda = 1$. We are interested in the left eigenvector.

✍ 3 Run *test_pr* + other drivers in matlab page

Kleinberg's Hubs and Authorities

- Idea is to put order into the web by ranking pages by their degree of Authority or "Hubness".
- An Authority is a page pointed to by many important pages.
  - Authority Weight = sum of Hub Weights from In-Links.
- A Hub is a page that points to many important pages:
  - Hub Weight = sum of Authority Weights from Out-Links.
- Source:

http://www.cs.cornell.edu/home/kleinber/auth.pdf

Computation of Hubs and Authorities

- Simplify computation by forcing sum of squares of weights to be 1.
- $\text{Auth}_j = x_j = \sum_{i:(i,j)\in \textbf{Edges}} \text{Hub}_i$.
- $\text{Hub}_i = y_i = \sum_{j:(i,j)\in \textbf{Edges}} \text{Auth}_j$.
- Let A = Adjacency matrix: $a_{ij} = 1$ if $(i,j) \in \textbf{Edges}$.
- $\mathbf{y} = A\mathbf{x}$, $\mathbf{x} = A^T\mathbf{y}$.
- Iterate . . . to leading eigenvectors of A^TA & AA^T.
- Answer: Leading Singular Vectors!

Centrality in graphs

- Goal: measure importance of a node, edge, subgraph, .. in a graph
- Many measures introduced over the years
- Early Work: Freeman '77 [introduced 3 measures] – based on 'paths in graph'
- Many different ways of defininf centrality! We will just see a few

Degree centrality: (simplest) 'Nodes with high degree are important'

$$C_D(v) = \deg(v)$$

Closeness centrality: 'Nodes that are close to many other nodes are important'

$$C_C(v) = \frac{1}{\sum_{w \neq v} d(v,w)}$$

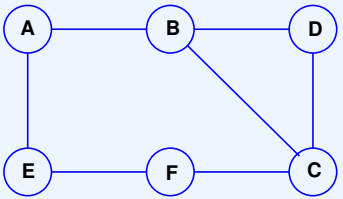
Betweenness centrality: (Freeman '77)

$$C_B(v) = \sum_{u \neq v, w \neq v} \frac{\sigma_{uw}(v)}{\sigma_{uw}}$$

- σ_{uw} = total # shortest paths from u to w
- $\sigma_{uw}(v)$ = total # shortest paths from u to w passing through v

✦ 'Nodes that are on many shortest paths are important'

Example: Find $C_D(v)$; $C_C(v)$; $C_B(v)$ when $v = C$



(u,w)	$\sigma_{uw}(v)$	σ_{uw}	/	(u,w)	$\sigma_{uw}(v)$	σ_{uw}	/
(A,B)	0	1	0	(B,E)	0	1	0
(A,D)	0	1	0	(B,F)	1	1	1
(A,E)	0	1	0	(D,E)	1	2	.5
(A,F)	0	1	0	(D,F)	1	1	1
(B,D)	0	1	0	(E,F)	0	1	0

- $C_D(v) = 3$;
- $C_C(v) = 1/[d_{CA} + d_{CB} + d_{CD} + d_{CE} + d_{CF}]$
  $= 1/[2 + 1 + 1 + 2 + 1] = 1/7$
- $C_B(v) = 2.5$ (add all ratios in table)

✍ 4 Redo this for $v = B$

Eigenvector centrality

- Supppose we have n nodes $v_j, j = 1, \cdots, n-$ each with a measure of importance ('prestige') p_j
- Principle: prestige of i depends on that of its neighbors.
- Prestige x_i = multiple of sum of prestiges of neighbors pointing to it

$$\lambda x_i = \sum_{j \in \mathcal{N}(i)} x_j = \sum_{j=1}^{n} a_{ji} x_j$$

- x_i = component of eigenvector associated with λ.
- Perron Frobenius theorem at play again: take largest eigenvalue $\rightarrow$ x_i's nonnegative